Warm-up

Compute the following limits:

$$(1)\ \lim_{x\to 0}\frac{x^3-x^2}{x+\sin(x)},\qquad (2)\ \lim_{x\to -1}\frac{x^2-2x-3}{x^2+5x+4},\qquad (3)\ \lim_{x\to 0^+} x\cot(x)$$

Medium

Compute the following limits:

$$(1)\ \lim_{x\to 1}\frac{\ln(x)}{\arcsin(2x-2)},\qquad (2)\ \lim_{x\to -\infty}\frac{e^x}{\arctan(x)+\pi/2},\qquad (3)\ \lim_{x\to\infty}(e^x+1)^{1/x},$$

$$(4)\ \lim_{x\to\infty}(20x)^{\frac{1}{\ln(5x)+1}},\qquad (5)\ \lim_{x\to\infty}\left(\frac{8x}{8x+10}\right)^{6x}.$$

Stretch

(1) Let

$$f(x)=\begin{cases} e^{-1/x^2} & x\neq 0\\ a & x=0\end{cases}$$

and determine the value of a that makes f continuous at 0.

(2) With f as defined above, compute $f'(0)$ by computing an appropriate limit:

$$f'(0)=\lim_{x\to 0}\frac{e^{-1/x^2}-a}{x-0}=\cdots$$